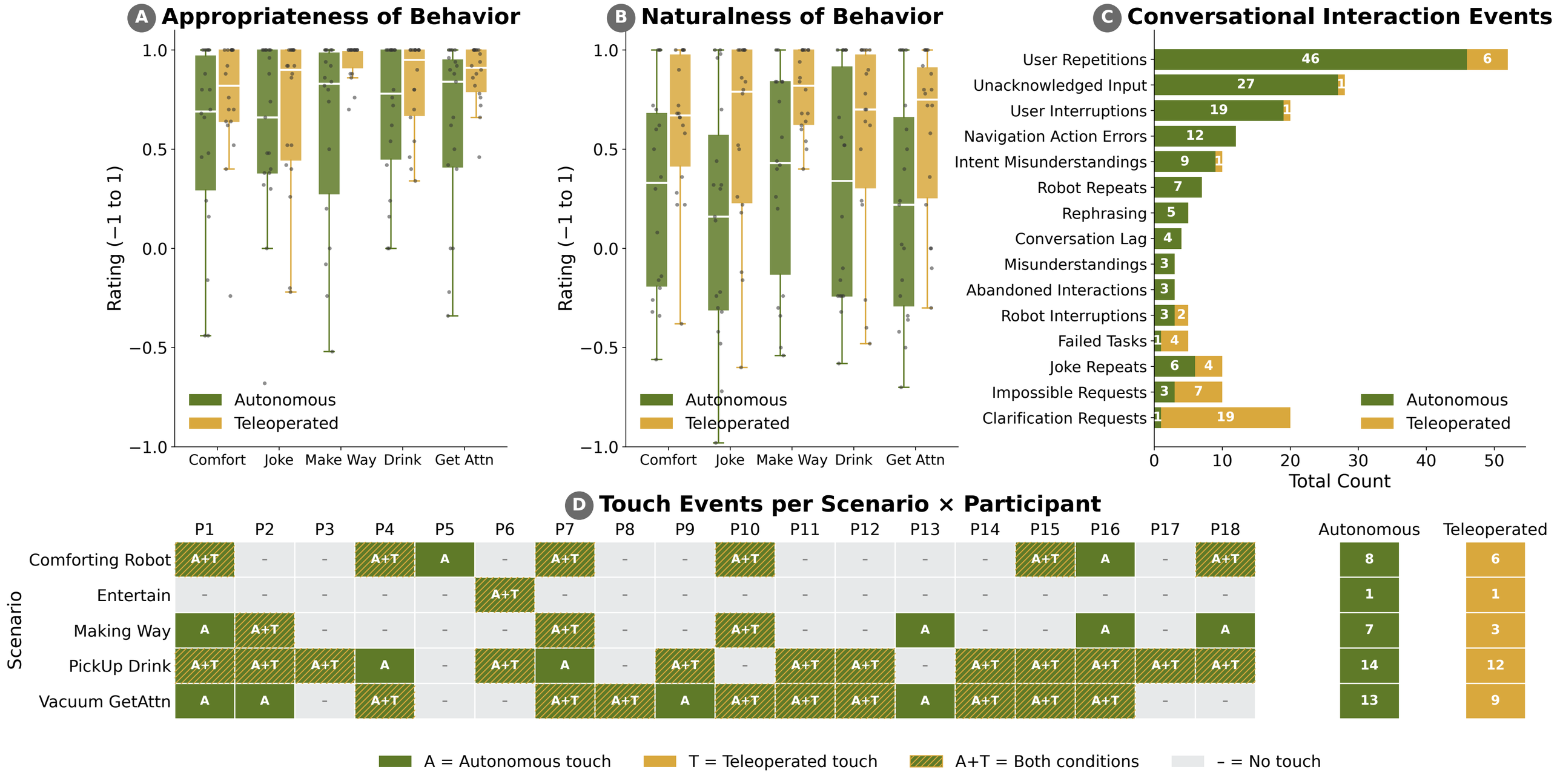
A Appropriateness of Behavior
Rating (−1 to 1)
1.0
0.5
0.0
−0.5
−1.0
Comfort
Joke
Make Way
Drink
Get Attn
Autonomous
Teleoperated
B Naturalness of Behavior
Rating (−1 to 1)
1.0
0.5
0.0
−0.5
−1.0
Comfort
Joke
Make Way
Drink
Get Attn
Autonomous
Teleoperated
C Conversational Interaction Events
User Repetitions 46 6
Unacknowledged Input 27 1
User Interruptions 19 1
Navigation Action Errors 12
Intent Misunderstandings 9 1
Robot Repeats 7
Rephrasing 5
Conversation Lag 4
Misunderstandings 3
Abandoned Interactions 3
Robot Interruptions 3 2
Failed Tasks 1 4
Joke Repeats 6 4
Impossible Requests 3 7
Clarification Requests 1 19
0
10
20
30
40
50
Total Count
Autonomous
Teleoperated
D Touch Events per Scenario × Participant
P1 P2 P3 P4 P5 P6 P7 P8 P9 P10 P11 P12 P13 P14 P15 P16 P17 P18
Autonomous
Teleoperated
Scenario
Comforting Robot A+T – – A+T A – A+T – – A+T – – – – A+T A – A+T 8 6
Entertain – – – – – A+T – – – – – – – – – – – – 1 1
Making Way A A+T – – – – A+T – – A+T – – A – – – A – A 7 3
PickUp Drink A+T A+T A+T A – A+T A – A+T – A+T A+T – A+T A+T A+T A+T A+T 14 12
Vacuum GetAttn A A – A+T – – A+T A+T A A+T A+T A+T A A+T A+T A+T – – 13 9
A = Autonomous touch
T = Teleoperated touch
A+T = Both conditions
– = No touch